MODEL CARD

Week 4 Day 1 — Breast Cancer Classification

1. Model Overview

A Random Forest classifier trained to classify breast cancer records using the scikit-learn breast cancer dataset. This model was developed for educational model-evaluation practice and is not approved for clinical use.

2. Intended Use

Educational experimentation with classification metrics, cross-validation and model evaluation. It must not be used to diagnose, screen, or make treatment decisions for patients.

3. Dataset

Dataset: scikit-learn Breast Cancer Wisconsin dataset. 569 records and 30 numeric features. Evaluation used a held-out test set and 5-fold stratified cross-validation.

4. Performance

Metric	Held-out test result
Accuracy	0.9561
Precision	0.9589
Recall	0.9722
F1 score	0.9655
ROC-AUC	0.9937
5-fold CV ROC-AUC (mean ± SD)	0.9885 ± 0.0083

5. Limitations and Risks

Performance metrics alone do not establish clinical safety, generalizability, calibration, or fairness across demographic groups. Dataset representativeness and subgroup performance require further investigation. Predictions must not replace professional medical assessment.

6. Responsible Use

Use only for learning and reproducible evaluation. Before any real-world deployment, conduct independent validation, subgroup fairness analysis, calibration checks, privacy review, and domain-expert assessment.

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